SMART INDIA HACKATHON 2024

- **Problem Statement ID – SIH1624**

- **Problem Statement Title- To develop an AI-based model for electricity demand projection including peak demand projection for Delhi power system .**

- **Theme- Smart Automation**

- **Category- Software**

- **Team ID-**

- **Team Name - *Load Logix AI***

AI MODEL FOR PREDICTING PEAK ELECTRICITY LOAD IN DELHI

Proposed Solution

We propose an AI-based predictive model that uses machine learning techniques to **predict peak electricity load** for the next day in Delhi. The model takes into account multiple factors such as **weather conditions, public holidays , solar generation data**.

How It Addressess the problem

Delhi faces major challenges in balancing electricity supply and demand due to **high load variability** between seasons and even within a day. Our solution helps predict peak loads accurately, enabling power distribution companies to **optimize power purchases and reduce costs**.

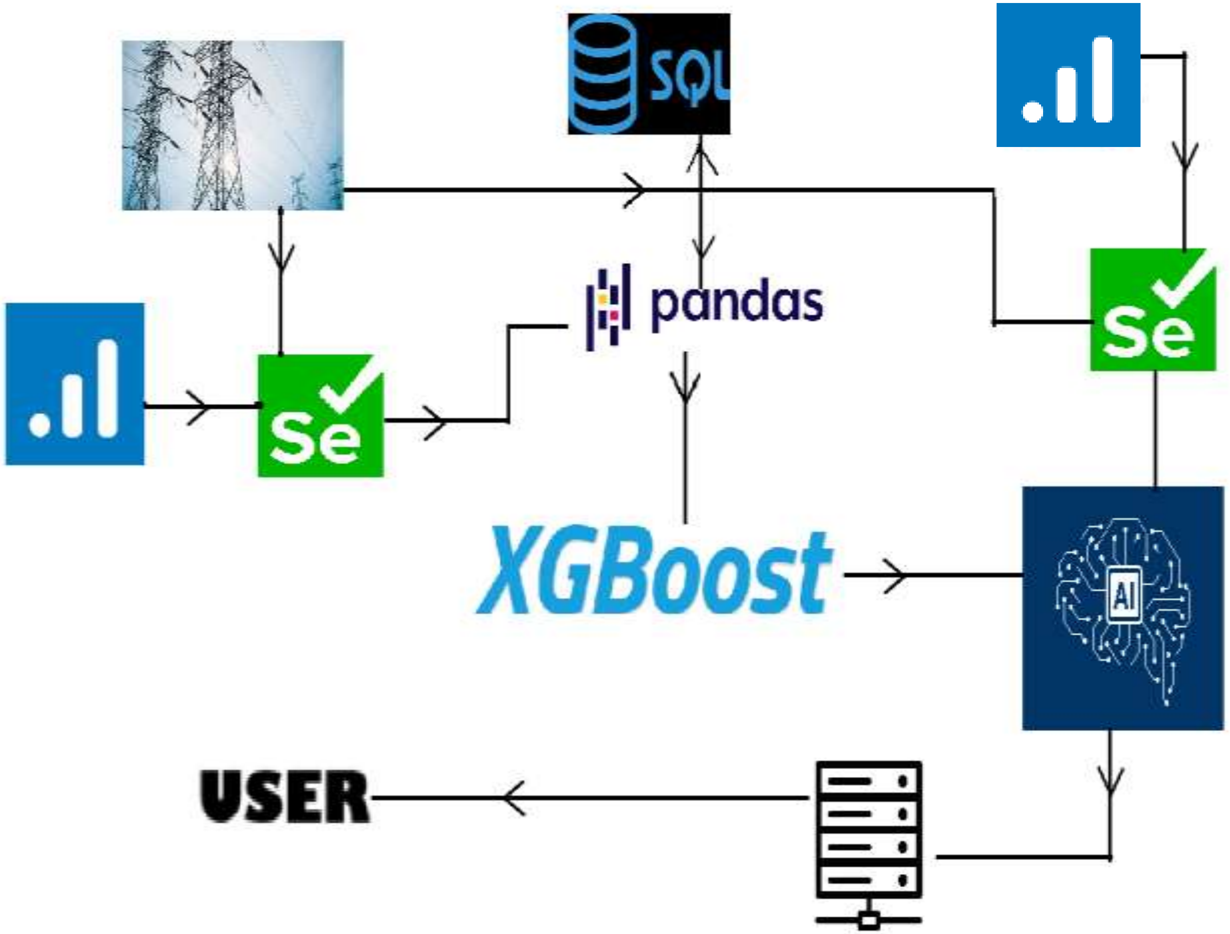
Innovation and Uniqueness

We incorporate **real-time weather data and public holiday** information to accurately capture load fluctuations.

The model also factors in the **duck curve effect** caused by solar generation, and the **uneven load growth** across different regions in Delhi.

TECHNICAL APPROACH

- **Data Collection**: Gathered **weather data** (temperature, humidity, wind speed), *load data from various regions, solar generation data, etc.*
- **Preprocessing:** Cleaned and preprocessed data by handling missing values, time-shifting features, and normalizing certain variables.
- **Modeling**: Implemented *a **Gradient Boosting Machine (GBM)** such as **XGBoost*** for predicting the peak electricity load. We used historical data to train the model.
- **Hyperparameter Tuning**: Used **manual tuning** and **GridSearchCV** to optimize the model's performance.
- **Evaluation**: The model's performance was measured using Root Mean Squared Error (RMSE), achieving an RMSE of **296 MW**.



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Feasibility and Viability

- The model is **technologically feasible** since it leverages existing data sources like weather forecasts and electrical load records. The model has been trained and tested on actual data from Delhi indicates strong predictive accuracy.

Potential Challenges and Risks

- **Data Limitations**: Limited historical data restrict model performance.
- **Real-time Data Availability**: Access to real-time weather and load data

Strategies

- **Data Augmentation**: Additional historical data can be sourced from utility providers or external datasets to further improve model accuracy.
- **Real-time Integration**: Collaborating with utility companies for continuous real-time data.

Impact and Benefits

- The model will help **optimize electricity procurement** in Delhi by accurately predicting peak loads, thus reducing the risk of power shortages or overspending.
- **Cost Savings**: Power distribution companies can make better purchasing decisions, reducing unnecessary expenses on excess power.
- **Grid Stability**: The model helps maintain **grid stability** by ensuring that sufficient power is available during peak times.
- ***Environmental Impact***: Reducing wastage of surplus power leads to a more efficient use of energy resources, minimizing the environmental footprint.
- **Scalability**: The model can be expanded to other regions facing similar electricity load prediction challenges and further can be implemented at microgrid and nanogrid levels too.

RESEARCH AND REFERENCES

Research

- **Gradient Boosting and XGBoost Models.**
- **Weather Data Impact on Electricity Load.**

References

- ***Application of AI Algorithms in Power System Load Forecasting Under the New Situation***.:*Jian Rong,Xiohua LIU,Kunyang GAO,State grid Hefei Power Supply Company.*
- **Meteostat .net** for Historical DATA of Weather and **Meteostat API** documentation for obtaining real-time weather data.
- **Delhi load dispatcher(delhisldc.org)** for Delhi's electricity and solar generation data.

Team Members

- **Team Leader: Amon Harsh**

Course:Btech Branch:CSE Year:2nd

Email:amonharsh11@gmail.com

- **Team Member: Gargi Priya**

Course:Btech Branch:CSE Year:2nd

Email:priya3480shine@gmail.com

- **Team Member: Ritik Raushan**

Course:Btech Branch:CSE Year:2nd

Email:ritikraushanbgs73@gmail.com

- **Team Member: Ansh Arya**

Course:Btech Branch:CSE Year:2nd

Email:ansharya927@gmail.com

- **Team Member: Kanika Singh**

Course:Btech Branch:ECE Year:2nd

Email:kanikasingh4231@gmail.com

- **Team Member: Sarvagya Dwivedi**

Course:Btech Branch:CSE Year:2nd

Email:sarvagyadwivedi0308@gmail.com